Naman Gupta

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Education

University of Southern California — Los Angeles, USA
MS in Computer Science — Jan 2026 – Dec 2027
Graphic Era Deemed to be University — Dehradun, India
B.Tech in Computer Science and Engineering — Sep 2021 – June 2025

Internship/Work Experience

Morgan Stanley — Mumbai, India

Technology Analyst — Aug 2025 – Nov 2025

- Diagnosed recurring performance bottlenecks across critical internal applications shared by multiple teams, then implemented targeted backend optimizations that measurably improved speed, stability, and day-to-day reliability.
- Partnered with engineering and operations stakeholders to design and build internal tools and reusable UI components for monitoring dashboards, improving operational visibility and cutting manual overhead across teams.

Technology Apprentice — July 2024 – Aug 2025

- Identified inefficiencies in manual, ad-hoc loan routing decisions and designed a data-driven recommendation model that surfaces the top 3 best-fit employees per department, cutting average task completion time by **20%**.
- Built and deployed predictive workload distribution models trained on historical throughput and staffing data, which raised balanced resource allocation accuracy across teams by **25%** while reducing individual employee overload.

Research Work

Deep Learning Approach for Malicious URL Detection | IEEE Paper — Feb. 2025

- Tasked with detecting malicious URLs at scale, trained and benchmarked four deep learning architectures (CNN, RNN, LSTM, Bi-LSTM) on a large IEEE-published dataset, achieving a top precision of **99.25%** with the Bi-LSTM model.
- Conducted a comparative architecture analysis for sequential pattern recognition in URL strings, showing that two-way context significantly boosts malicious URL classification accuracy and generalization performance.

Technical Skills

Languages: C, C++, Java, JavaScript, Python, HTML
Developer Tools: VS Code, Git, Microsoft, PyCharm, Jupyter Notebook
Framework / Library: Node.js, ReactJS, Flask, Django, TensorFlow, Keras, Pandas

Projects

TechHire: AI Job Search Platform GitHub | *Python, FastAPI, React, PostgreSQL* — May 2026 – June 2026

- Built a full-stack tech job platform with a Python scraper and FastAPI backend, enabling fast search and filtering across postings by skills, salary range, visa sponsorship, and remote or hybrid work mode.
- Integrated a multi-model Groq LLM pipeline that synthesizes job summaries from three models and built an AI resume analyzer that scores candidate fit and rewrites weak bullets against the target job description.

Mind Map: Mental Wellbeing Detection GitHub | *Python, BERT, LLaMA, Web Scraping* — Dec 2024 - June 2025

- Set out to flag early signs of mental health decline within online communities, web-scraped large volumes of Reddit posts and comments, then cleaned and structured the data for downstream sentiment analysis pipelines.
- Fine-tuned BERT and LLaMA models to classify user sentiment and surface early indicators of mental wellbeing concerns, enabling automated screening across thousands of scraped posts in real time.

Real-Time Sign Language Detection GitHub | *Python, OpenCV, TensorFlow, Keras* — Apr. 2024 – May 2024

- Set out to make sign language communication more accessible, built a real-time computer vision pipeline using OpenCV to capture and preprocess live hand gesture video streams from a webcam.
- Trained and fine-tuned a TensorFlow/Keras deep learning model to classify and translate hand gestures into text in real time, achieving an overall accuracy of **98%** in live testing.

A full-featured Blogging Website GitHub | *Flask, SQLAlchemy, Python, Bootstrap* — Nov. 2023

- Designed and built a full-featured blogging platform end-to-end using Flask and SQLAlchemy, implementing user authentication, post creation, and a relational database schema from scratch.
- Styled a responsive, mobile-friendly interface with Bootstrap and iterated on the UX based on user feedback, which drove a **20%** increase in overall user engagement and satisfaction.